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ARTIFICIAL NEURAL NETWORK

Abstract

A neural network system is provided. The neural network system may include an input layer configured to receive an input signal. The neural network system may also include a first set of weights downstream of the input layer. The neural network system may further include a second set of weights downstream of the input layer, the second set of weights being connected by a mathematical relationship to the first set of weights. The neural network system may additionally include an output layer, downstream of the first set of weights and the second set of weights and configured to generate an output signal in response to the input signal.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is related to U.S. Pat. No. 10,902,914, entitled “Programmable resistive memory element and a method of making the same,” filed Jun. 4, 2019, and issued Jan. 26, 2021, which is hereby incorporated by reference in its entirety. This application is also related to U.S. Pat. No. 11,183,240, entitled “Programmable resistive memory element and a method of making the same,” filed Jan. 26, 2021, and issued Nov. 23, 2021, which is also hereby incorporated by reference in its entirety. This application is also related to U.S. patent application Ser. No. 18/048,594, entitled “Analog programmable resistive memory,” filed Oct. 21, 2022, which is also hereby incorporated by reference in its entirety.

BACKGROUND

[0002] An artificial neural network is a computational model that mimics the behavior of the human brain. Such a network learns from data, and, based on the learning, predicts or recognizes specific patterns. To obtain a network with these properties, large quantity of data is needed for training and analysis, which consumes a large amount of computational power. With the advent of new algorithms that are more power-hungry, the need to find more efficient networks is highly desirable. One solution is to construct such networks at the hardware level, but such networks have so far proven complex to adapt to the flexibility afforded by networks constructed at the software level. For example, negative weights are difficult to implement in hardware-based neural networks. This difficulty can be avoided by implementing some steps at the hardware level and others at the software level. But such implementations using conventional techniques cause undesirable latency.

[0003] As such, a significant improvement in neural networks and neural network systems is desired.

SUMMARY

[0004] Embodiments disclosed herein solve the aforementioned technical problems and may provide other solutions as well. In one or more embodiments, an artificial neural network system is formed with at least two sets of weights for each layer, the two sets of weights being connected by a mathematical relationship involving one input layer and one output layer The input layer consists of any type of pattern, including but not limited to, image patterns, video patterns, audio patterns, text patterns, and/or any type of organized information, in raw format or preprocessed through a mathematical transformation of the raw pattern. In one embodiment, the network output layer is formed by neurons having two components, one behaving like an excitatory one and another behaving like an inhibitory one. In another embodiment, the network the output layer contains two neurons, each neuron having two components, one behaving like an excitatory one and another behaving like an inhibitory one.

[0005] In another embodiment, the values of the weights of the artificial neural network may be adjusted via a suitable algorithm. For example, an input pattern may be sent via the input layer and the output that is produced by that pattern may be compared with the desired output with a suitable algorithm. Then, the weights for each input neuron can be adjusted according to the algorithm such that the system would eventually identify the desired pattern. In other embodiments, the weights can be restricted to be positive values. In other embodiments, the weights can be further restricted to be between 0 and 1 such that it is easier to map the values of a memristor crossbar.

[0006] In some other embodiments, the artificial neural network is formed with at least one hidden layer, for implementing more complex behavior. In such cases, the artificial neural network may be able to identify more complex patterns, predict and learn from large quantities of data. In such cases, the internal structure of a hidden layer may be similar to the structure of the output layer, but the hidden layer being able to identify certain features of a pattern such that the system would identify easier the desired pattern.

[0007] In another embodiment, the artificial neural network can be implemented at the software level as well as at the hardware level. When implemented at the hardware level, the system comprises of at least one memristor crossbar and artificial neurons built with electronic components. In one or more embodiments, the values of the resistances of the memristors from the crossbar may be modified to represent the weights of an artificial neural network trained in advanced. In other embodiments, the training of the weights can be done directly on the hardware artificial neural network. For example, an input pattern may be sent via the input layer and the output that is produced by that pattern may be compared with the desired output with a suitable algorithm. Then, the values of the resistances can be adjusted accordingly to the algorithm such that the system would eventually identify the desired pattern.

[0008] In one or more embodiments, a neural network system may be provided. The neural network system may include an input layer configured to receive an input signal and a first set of weights and a second set of weights downstream of the input layer. The second set of weights may be connected by a mathematical relationship with the first set of weights. The neural network system may further include an output layer downstream of the first set of weights and the second set of weights. The output layer may be configured to generate an output signal in response to the input signal. In some embodiments, the output layer may be configured to act as a hidden layer in the network, its output signal acting as the input signal for the next layer in the network.

[0009] In one or more embodiments, a method of training a memristors based neural network system is provided. The method may include providing inputs to a software system and retrieving corresponding outputs. The method may further include comparing the corresponding outputs to an expected output. The method may also include adjusting weights in the software system until the corresponding outputs are within a predetermined distance of an expected output. The method may additionally include. mapping the weights to a memristor crossbar of the memristors based neural network system.

[0010] In one or more embodiments, a method of training a memristors based neural network system is provided. The method may include providing inputs to the memristors based neural network system and retrieving corresponding outputs. The method may also include comparing the corresponding outputs to an expected output. The method may further include adjusting resistances of a memristor crossbar of the memristors based neural network system until the corresponding outputs are within a predetermined distance of the expected output.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. **1** depicts an example of a neural network system, based on the principles disclosed herein.

[0012] FIG. **2** depicts an example of a neural network system, based on the principles disclosed herein.

[0013] FIG. **3** depicts an example of a neural network system, based on the principles disclosed herein.

[0014] FIG. **4** depicts an example of a neural network system, based on the principles disclosed herein

[0015] FIG. **5** depicts an example of an internal structure of a hidden layer of a neural network system, based on the principles disclosed herein.

[0016] FIG. **6** depicts a flow diagram of an example method of training a memristors based neural network system, based on the principles disclosed herein.

[0017] FIG. **7** depicts a flow diagram of an example method of training a memristors based neural network system, based on the principles disclosed herein.

DETAILED DESCRIPTION

[0018] FIG. **1** depicts an example of a neural network system **100**, based on the principles disclosed herein. It should be understood that the neural network system **100** is provided as an example and should not be considered limiting. That is, neural network systems with additional, alternative, or fewer components should be considered within the scope of this disclosure. The neural network system **100** may form one or more layers of a larger, neural network-based system. The neural network system **100** may be implemented on software and/or hardware levels.

[0019] As shown, the neural network system **100** incorporates a layer of weights **101** comprising a first set of weights (W) **102** converging on a neuron **112** and a second set of weights (1-W) **103** converging on a neuron **113**. That is, the first set of weights **102** are complementary to the second set of weights **103** (i.e., W and 1-W complement each other). The sets of weights **102**, **103** are formed by corresponding connections **111**. The neurons **112**, **113** perform mathematical operations based on the sets of weights **102**, **103**, respectively, and corresponding biases (b) **117**. The neural network system **100** further incorporates an input layer **104** formed by input neurons **114** and output layer **105** formed by output neurons **115**, **116**. The output neurons **115**, **116** interface with the neurons **112**, **113**.

[0020] The input layer **104** may receive any type of pattern, including but not limited to, image patterns, video patterns, audio patterns, text patterns, and/or any type of organized information. In one or more embodiments, the input may be in raw format. In one or more embodiments, the input may be mathematical transformation of the raw format. The output layer **105** may generate a label for the received pattern, where the label indicates an identification of that specific pattern by the neural network system **100**. In one or more embodiments, the input may be in a form of an input signal.

[0021] In one or more embodiments the layer of weights **101** could be trained using an algorithm, e.g., backpropagation, for identifying a pattern from a set of inputs (e.g., images). By inputting a pattern via the input layer **104** and comparing the output of neural network system **100** with the desired label, the weights **101** can be adjusted, e.g., by backpropagating the errors, such that the neural network system **100** learns to eventually identify the pattern. It should be understood that backpropagation is just an example, and any algorithm that adjusts the layer of weights **101** such that the neural network system **100** may be able to predict, identify or learn from the input data/pattern could be used.

[0022] The weights **102**, **103** may have positive or negative values. In some embodiments with hardware implementations, the weights **102**, **103** may have only positive values. One or more values of the weights **102**, **103** may be scaled between a minimum value and a maximum value. In some embodiments, the minimum value could be 0 and the maximum value could be 1. Such scaling may facilitate the mapping of the weights **102**, **103** on a hardware system as described with reference to FIG. **4** below.

[0023] In one or more embodiments, the neurons **112**, **113** may perform the summing of the weights **102**, **103** multiplied by the input neurons **114** and the corresponding biases **117**. That is, as an example of applying mathematical operations described above, the neuron **112** may be updated by the sum of the weights **102** multiplied by the input neurons **114**, and the neuron **113** may be the sum of the weights **103** multiplied by the input neurons **114**. And, the corresponding biases **117** may be added to each of the neurons **112**, **113**.

[0024] As such, an activation function may be performed with the neurons **112**,**113** to obtain the corresponding values in the output neurons **115**, **116** forming the output layer **105** downstream of the input layer **104**. In one or more embodiments, the neuron **112** may form an excitatory component and the neuron **113** may form an inhibitory component for a first activation function for the output neuron **115**. In one or more embodiments, the neuron **112** may form an inhibitory component and the neuron **113** may form an excitatory component for a second activation function for the output neuron **116**. In one or more embodiments, the value of the output neuron **116** may be complementary to the value of the output neuron **115**. In one or more embodiments, the outputs may be in the form output signals.

[0025] FIG. **2** depicts an example of a neural network system **200**, based on the principles disclosed herein. The neural network system **200** may be implemented at software and/or hardware levels. As shown, the neural network system **200** incorporates two layers of weights **201** each comprising a first set of weights (e.g., W) **202**, and a second set of weights (e.g., 1-W) **203**, both of which may be formed by corresponding connections **211**. The neural network system **200** incorporates input layer **204** formed by input neurons **214** and output layer **205** formed by output neurons **215**,**216**. The neural network system **200** comprises a hidden layer **206** formed by hidden neurons **225**,**226**.

[0026] In one or more embodiments, the neural network system **200** may be formed by incorporating two neural network systems **100**, with each constituent neural network system **100** having different trained weights for recognizing a pattern and with different sizes of the layers. For example, the system formed by input layer **204**, the set of weights **201**, the summing neurons **212**, **213** and the hidden layer **206** (forming an output layer) may be considered a first system **100**. The system formed by the hidden layer **206**, the set of weights **201**, and the output layer **205** with their corresponding summing neurons may form another system **100**, where the input layer is formed by the hidden layer **206**.

[0027] In this setup, the input neurons **214** may receive any type of pattern, including but not limited to, image patterns, video patterns, audio patterns, text patterns, and/or any type of organized information. In one or more embodiments, the input may be in raw format. In one or more embodiments, the input may be a mathematical transformation of the raw format. In one or more embodiments, the input may be in a form of an input signal. The hidden layer **206** may learn features that would assist the neural network system **200** to identify an input pattern. As shown, the hidden layer **206** may include a hidden neuron **225** that may be connected to the neurons **212**, **213**. In one or more embodiments, the neuron **212** may form an excitatory component while the neuron **213** may form an inhibitory component for the hidden neuron **225**. But for another hidden neuron **226**, the neuron **212** may form an inhibitory component, while the neuron **213** may form an excitatory component.

[0028] As shown, the neuron **212** may calculate the sum of the input neurons **214** multiplied by the first set of weights **202**, and the neuron **213** may calculate the sum of the input neurons **214** multiplied by the set of weights **203**. Corresponding biases **217** may be added to both the neurons **212**, **213**. An activation function may be performed with the neurons **212**, **213** to obtain corresponding outputs to the neurons **225**, **226** of the hidden layer **206**. In one or more embodiments, the value of the output neuron **226** may be complementary to the value of the output neuron **225**. In one or more embodiments, the outputs may in a form of output signals.

[0029] The layer of weights **201** may include, but is not limited to, positive values. The layer of weights **201** may be further scaled between 0 and 1. Such scaling may facilitate the mapping the layer of weights **201** on a hardware system as described with reference to FIG. **4** below.

[0030] The output layer **205**, comprising the output neurons **215**, **216**, may generate a label for the received pattern, where the label indicates an identification of that specific pattern by the neural network system **200**. The result of the output neuron **215** may be determined by connected, upstream components of the neural network system **200**. As such, a complex function may be formed by adding non-linearity to the neural network system **200** by using the hidden layer **206** with an activation function and/or mathematical rules performed over the neurons **212**, **213**. In one or more embodiments, the value of the output neuron **216** is complementary to the value of the output neuron **215**.

[0031] FIG. **3** depicts an example of a neural network system **300**, based on the principles disclosed herein. The neural network system **300** may be implemented at hardware and/or software levels. As shown, the neural network system **300** incorporates multiple layers of weights **301**, each comprising a first set of weights (W) **302**, and a set of weights (1-W) **303**, where the individual weights may be formed by connections **311**. The neural network system **300** incorporates an input layer **304** formed by input neurons **314** and an output layer **305** formed by output neurons **315**, **316**.

The neural network further comprises multiple hidden layers **306** formed by hidden neurons **325**, **326** which perform mathematical operations on the first set of weights **302** and for the second set of weights **303**, similar to the method of calculating neurons **226** and **225**, as shown in FIG. **2**. The mathematical operations may further include the corresponding biases **317**.

[0032] It should be noted that the neural network system **300** may include any number of input neurons (e.g., input neuron **314**), any number of hidden layers (e.g., hidden layer **306**) with any number of hidden neurons (e.g., hidden neurons **325**, **326**), and any number of output neurons (e.g., output neurons **315**, **316**).

[0033] An input layer **304** may receive any type of pattern, including but not limited to, image patterns, video patterns, audio patterns, text patterns, and/or any type of organized information. In one or more embodiments, the input may be in a raw format. In one or more embodiments, the input may be a mathematical transformation of the raw format. In one or more embodiments, the input may in a form of an input signal.

[0034] The hidden layer **306** (i.e., multiple hidden layers) may learn features of patterns that may assist the neural network system **300** to identify a specific pattern during deployment. As shown, the hidden layer **306** may include a hidden neuron **325** that may be composed of neurons **312**, **313**. In one or more embodiments, the neuron **312** may form an excitatory component while the neuron **313** may form an inhibitory component. But for another hidden neuron **326**, the neuron **312** may form an inhibitory component, while the neuron **313** may form an excitatory component.

[0035] As shown, the neuron **312** may calculate the sum of the input neurons **314** multiplied by the first set of weights **302**, and the neuron **313** may calculate the sum of the input neurons **314** multiplied by the second set of weights **303**, for the first hidden layer. For the other hidden layers, the neuron **312** may calculate the sum of the output neurons of the previous hidden layer multiplied by the first set of weights **302**, and the neuron **313** may calculate the sum of the output neurons of the previous hidden layer multiplied by the second set of weights **303**. Corresponding biases may be added to both the neurons **312**,**313**. An activation function may be performed with the neurons **312**, **313** to obtain corresponding outputs to the neurons **325**, **326** of the hidden layer **306**. In one or more embodiments, the value of the neuron **326** is complementary to the value of the neuron **325**.

[0036] The output layer **305**, comprising the output neurons **315**, **316**, may generate a label for the received pattern, where the label indicates an identification of that specific pattern identified by the neural network system **300**. The result of the output neuron **315**, **316** may be determined by connected, upstream components of the neural network system **300**. As such, a complex function may be formed by adding non-linearity to the neural network system **200** by using the hidden layers **306** with an activation function and/or mathematical rules performed over the neurons **312**, **313**. In one or more embodiments, the value of the output neuron **316** is complementary to the value of the output neuron **315**. In one or more embodiments, the outputs may be in the form of output signals.

[0037] It should be noted that other types of architectures with any number of neurons that can be implemented using a first set of weights **302** and a second set of weights **303**, connected to the first set of weights **302** by a mathematical operation (also referred to as a mathematical relationship) should be considered under the scope of this disclosure. In one or more embodiments, the mathematical relationship between the set of weights may be W and 1-W.

[0038] FIG. **4** depicts an example of a neural network system **400**, based on the principles disclosed herein. For instance, the neural network system **400** may be a hardware level implementation of the neural network system **200** shown in FIG. **2**. It should be understood that the neural network system **400** is provided as example and should not be considered limiting. That is, neural network systems with additional, alternative, or fewer number of components, should be considered within the scope of this disclosure. The neural network system **400**, using the principles disclosed herein, may form one or more layers of a larger hardware based neural. For the hardware-based neural network embodiments, the training may adjust network weights (e.g., formed by

memristors).
[0039] As shown, the neural network system **400** may include a memristor crossbar **401** formed with column **402** for implementing a first set of weights W, and with column **403** for implementing a second set of weights 1-W that complement the first set of weights W. The columns **402**, **403** may be formed memristors **411**. The memristor crossbar **401** could contain any kind of memristors, or structures combining memristor with other elements like diodes, transistors, and/or any other type of electronic components. In one or more embodiments, the memristors **411** may include indium gallium zinc oxide (IGZO) memristors, having electrodes situated on a same plane.
[0040] The neural network system **400** may further include a bias **417**, an input layer **404** formed by input neurons **414**, an output layer **405** formed by output neurons **415**, **416**, and an indicator **418**. The neural network system **400** may further include a hidden layer **406**, formed by hidden neurons **425**, **426**, and connections **412** for the first set of weights W and connections **413** for the complementary second set of weights 1-W.
[0041] The input layer **404** may receive any type of pattern, including but not limited to, image patterns, video patterns, audio patterns, text patterns, and/or any type of organized information. In one or more embodiments, the input may be in a raw format. In one or more embodiments, the input may be a mathematical transformation of the raw format. The output layer **405** may include output neurons **415**, **416** that may generate any type of signal/trigger (and/or a label) identifying a specific pattern sent by the input layer **404**. An indicator **418** may indicate the triggering of the output neuron **416**. In one or more embodiments, the value of the output neuron **416** is complementary to the value of the output neuron **415**. In one or more embodiments, the value of the output neurons **415**, **416** may be dependent on the other upstream, connected components of the neural network system **400**.
[0042] The connection **412** may perform the summing of the input layer **404**, where each input neuron **414** may be multiplied with a weight from the first set of weights W (as represented in the column **402**), while connection **413** may perform the summing of the input layer **404**, where each input neuron **414** may be multiplied with a weight from the second set of weights 1-W (as represented in the column **403**). To both sums, corresponding biases **417** may be added. In one or more embodiments, the biases may be implemented at the hardware level by adding another row of synaptic weights and applying a constant input on that row.
[0043] In one or more embodiments, the resistance values of the memristors **411** from the memristor crossbar **401** may be mapped from a software based neural network system. In one or more embodiments, the resistance values of the memristors **411** may be subsequently updated after each iteration of a training algorithm. For the training, any type of algorithm that is configured to adjusts the weights—e.g., by changing the resistance values of the memristors **411**—can be used. It should be noted that the modifying of the resistance values of memristors **411** may be performed by any type of control, regardless of whether the resistance values are mapped to software for the training or whether the training is performed at the hardware level directly. It should be also noted that such resistance values may be configured during a fabrication of the memristor crossbar **401**, e.g., in the cases of mapping the values from software.
[0044] In one or more embodiments, an output value of a connection **412** may form an excitatory component while the output value of the connection **413** may form an inhibitory component of neuron **426**. In one or more embodiments, the output value of the connection **412** may form an inhibitory component while the output value of the connection **413** may form an excitatory component for the neuron **425**. The output values of the neuron **425**, **426** may depend on the output values of the connections **412**,**413** over which a mathematical rule, such as an activation function, was performed.
[0045] The hidden layer **406**, comprising the hidden neurons **416**, may improve the performance of the neural network system **400** by adding complexity and non-linearity. For example, the hidden neurons **425**, **426** may learn features of the input pattern that may facilitate the neural network

system **400** to identify that input pattern.

[0046] It should be noted that other types of architectures with any number of neurons that can be implemented using a first set of weights in the column **402** and a second set of weights in the column **403**, connected to the first set of weights by a mathematical operation (also referred to as mathematical relationship) should be considered under the scope of this disclosure. In one or more embodiments, the mathematical relationship between the set of weights may be W and 1-W.

[0047] FIG. **5** depicts an example of an internal structure of a hidden layer **500** (e.g., hidden layer **406** shown in FIG. **4**), based on the principles disclosed herein. The hidden layer may comprise neurons **525**, **526**, and current summing neurons **512**, **513**. In one or more embodiments, the hidden layer **500** may be implemented at the hardware level. It should be noted that this is just an example of an internal structure of the hidden layer **500** and other structures—with additional, alternate, or fewer number of components—may be used.

[0048] As shown, a neuron **525** comprises resistances **532**, **535**, transistors **533**, **531**, **538**, ground references **534**, **536**, and voltage references **539**, **537**. A neuron **526** comprises resistances **542**, **545**, transistors **543**, **541**, **548**, ground references **544**, **546**, and voltage references **549**, **547**. The current summing neuron **512** comprises a resistance **552**, a ground reference **555**, and a wire node **550**. The current summing neuron **513** comprises a resistance **553**, a ground reference **554**, and a wire node **551**.

[0049] The neurons **525**, **526** may perform a mathematical rule (or operation) over the current summing neurons **512**, **513**, respectively. In one or more embodiments, the neuron **525** may consider the current summing neuron **513** as an excitatory component and the current summing neuron **512** as an inhibitory component, while the neuron **526** may consider the current summing neuron **513** as an inhibitory component and the current summing neuron **512** as an excitatory component.

[0050] In one or more embodiments, the value from the current summing neuron **512** may be transmitted on transistors **541**, **533**, while the values from the current summing neuron **513** may be transmitted on transistors **543**.**531**. The transistor **543**, together with the resistor **542** and ground **544**, may form the inhibitory component and the transistors **541**, **548** together with the resistor **545**, ground **546** and voltage references **547**,**549**, may form the excitatory component of the neuron **526**, while the transistor **533**, together with the resistor **532** and ground **534**, may form the inhibitory component and the transistors **531**, **538** together with the resistor **535**, ground **536** and voltage references **537**,**539**, may form the excitatory component of the neuron **525**.

[0051] In one or more embodiments, the neuron **526** may be a complement of the neuron **525**, e.g., when one is turned off, the other is turned on. This complementarity may be achieved because if the value of the current summing neuron **512** is high enough (e.g., above a threshold), then the transistor **533**, using the resistance **532**, may inhibit the transistor **531**, such that the transistor **538** would not be open, causing the neuron **525** to be turned off, but the transistor **541** may be open, therefore, the transistor **548** would be open, causing the neuron **526** to be turned on. Similarly, when the value of the current summing neuron **513** is high enough, the transistor **543** using the resistance **542** may inhibit transistor **541**, such that the transistor **548** would not be open, causing the neuron **526** to be turned off, but the transistor **531** may be open, and therefore, the transistor **538** would be open, causing the neuron **525** to be turned on.

[0052] The ground references **534**, **544**, **536**, **546**, and voltage references **537**, **547**, **539**, **549**, and resistances **535**, **545** may allow to adjust the functionality of a neuron (e.g., neurons **525**, **526**) as a whole.

[0053] The functionality of these transistors may depend on their type and the values from the current summing neurons **512**, **513**. For instance, the transistors **543**, **541**, **548** combinedly may function as an activation function for the neuron **526**. Similarly, the transistors **533**, **531**, **538** combinedly may function as an activation function for the neuron **525**. The corresponding activation functions may achieved by performing a mathematical rules with the current summing neurons **512**, **513**.

[0054] In one or more embodiments, a layer of hidden neurons, such as those in the hidden layer **500**, may be also used for an output layer. Such output layer may include an indicator similar to **418**, which may indicate that the corresponding neural network system has recognized the pattern. In one or more embodiments, the neuron **525** may identify a pattern while the neuron **526** may identify the inverse of a pattern.

[0055] It should be noted that this type of artificial neuron should not be considered limiting and other neurons with fewer, more, or other types of components and or functionalities should be considered within the scope of this disclosure.

[0056] FIG. **6** depicts a flow diagram of an example method **600** of training a memristors based neural network system, based on the principles disclosed herein. The method **600** may follow a software based approach of training the memristors based neural network system. It should be understood that the steps of the method **600** are just examples and not limiting. Therefore, methods with additional, alternative, or additional number of steps should be considered within the scope of this disclosure.

[0057] The method may begin at step **602**, where inputs are provided to a software system and corresponding outputs are retrieved. The software system may comprise a software analogue of the memristors based neural network system where the individual weights may be initialized and subsequently adjusted in software.

[0058] At step **604**, the corresponding outputs may be compared with an expected output. The expected output may represent the correct output and the training steps may have to lead to the software system approaching the correct output.

[0059] At step **606**, the weights of the software system may be adjusted until the corresponding outputs are within a predetermined distance of the expected output. The adjustment can be performed using any type of training algorithm, such as backpropagation.

[0060] At step **608**, the weights of the software system may be mapped to a memristor crossbar of the memristors based neural network system. The mapping may include adjusting the resistances of individual memristors within the memristor crossbar to correspond to the weights of the software system. After the mapping, the memristors based neural network system is trained and ready for testing.

[0061] FIG. **7** depicts a flow diagram of an example method **700** of training a memristors based neural network system, based on the principles disclosed herein. The method **700** may follow a hardware based approach of training the memristors based neural network system. It should be understood that the steps of the method **700** are just examples and not limiting. Therefore, methods with additional, alternative, or additional number of steps should be considered within the scope of this disclosure.

[0062] The method may begin at step **702**, where inputs are provided to the memristors based neural network system and corresponding outputs are retrieved. The corresponding outputs may be determined by the current resistance values within the memristor crossbar. In the subsequent steps, the resistance values may be changed iteratively.

[0063] At step **704**, the corresponding outputs are compared to an expected output. The expected output may represent the correct output and the training steps may lead the neural network system approaching the correct output.

[0064] At step **706**, the resistance values within the memristor crossbar are adjusted until the corresponding outputs are within a predetermined distance of the expected output. Once this condition is reached, the memristors based neural network system is trained and ready for testing.

[0065] Additional examples of the presently described method and device embodiments are suggested according to the structures and techniques described herein. Other non-limiting examples may be configured to operate separately or can be combined in any permutation or combination with any one or more of the other examples provided above or throughout the present disclosure.

[0066] It will be appreciated by those skilled in the art that the present disclosure can be embodied in other specific forms without departing from the spirit or essential characteristics thereof. The presently disclosed embodiments are therefore considered in all respects to be illustrative and not restricted. The scope of the disclosure is indicated by the appended claims rather than the foregoing description and all changes that come within the meaning and range and equivalence thereof are intended to be embraced therein.
[0067] It should be noted that the terms “including” and “comprising” should be interpreted as meaning “including, but not limited to”. If not already set forth explicitly in the claims, the term “a” should be interpreted as “at least one” and “the”, “said”, etc. should be interpreted as “the at least one”, “said at least one”, etc. Furthermore, it is the Applicant's intent that only claims that include the express language “means for” or “step for” be interpreted under 35 U.S.C. 112(f). Claims that do not expressly include the phrase “means for” or “step for” are not to be interpreted under 35 U.S.C. 112(f).

Claims

1. A neural network system comprising: an input layer configured to receive an input signal; a first set of weights and a second set of weights, both downstream of the input layer, the second set of weights being connected by a mathematical relationship with the first set of weights; and an output layer, downstream of the first set of weights and the second set of weights, and configured to generate an output signal in response to the input signal.
2. The neural network system of claim 1, comprising: one or more hidden layers, each hidden layer being formed by one or more neurons, each hidden layer comprising a first component and a second component, the first component configured to function as an excitatory component and the second component configured to function as an inhibitory component.
3. The neural network system of claim 1, the output layer comprising one or more neurons, each comprising a first component and a second component, the first component configured to function as an excitatory component and the second component configured to function as an inhibitory component.
4. The neural network system of claim 3, a first neuron of the one or more neurons being configured to detect an input pattern and a second neuron of the one or more neurons being configured to detect an inverse of the input pattern.
5. The neural network system of claim 1, the mathematical relationship comprising at least one weight in the first set of weights having a value W and at least one weight in the second set of weights having a value 1-W.
6. The neural network system of claim 1, the first set of weights and the second set of weights comprising positive values.
7. The neural network system of claim 1, the first set of weights and the second set of weights comprising values scaled between 0 and 1.
8. The neural network system of claim 1, comprising: at least one memristor crossbar comprising a plurality of memristors and implementing the first set of weights and the second set of weights.
9. The neural network system of claim 8, the plurality of memristors comprising Indium gallium zinc oxide (IGZO) based memristors.
10. The neural network system of claim 8, the plurality of memristors having corresponding electrodes situated on the same plane.
11. The neural network system of claim 8, comprising neurons being formed with electronic components.
12. The neural network system of claim 11, the neurons each comprising two components, a first component configured to function as an excitatory portion and a second component configured to function as an inhibitory portion.

13. The neural network system of claim 8, comprising at least a hidden layer being formed by neurons with electronic components.
14. The neural network system of claim 13, the neurons each comprising two components, a first component configured function as an excitatory portion and a second component configured to function as an inhibitory portion.
15. A neural network system of claim 13, the output layer comprising a first neuron and a second neuron, the first neuron configured to identify a pattern from the input signal and the second neuron configured to identify an inverse of the pattern.
16. A neural network system of claim 8, the first set of weights having a value W and the second set of weights having a complementary value 1-W.
17. A method of training a memristors based neural network system, the method comprising: providing inputs to a software system and retrieving corresponding outputs; comparing the corresponding outputs to an expected output; adjusting weights in the software system until the corresponding outputs are within a predetermined distance of the expected output; and mapping the weights to a memristor crossbar of the memristors based neural network system.
18. The method of claim 17, further comprising: testing the memristors based neural network system by providing additional inputs.
19. A method of training a memristors based neural network system, the method comprising: providing inputs to the memristors based neural network system and retrieving corresponding outputs; comparing the corresponding outputs to an expected output; and adjusting resistances of a memristor crossbar of the memristors based neural network system until the corresponding outputs are within a predetermined distance of the expected output.
20. The method of claim 19, further comprising: testing the memristors based neural network system by providing additional inputs.